

SHEETAL

Sustainable Hydrogel-Enabled Energy Efficient Thermal
Augmentation for Air-conditioning Load Reduction

Indian innovation for energy-efficient cooling.

Team MEGHDOOT

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The Cooling Problem at IIT Kharagpur

1000+

Split AC units
across IIT KGP campus

Faculty, labs, hostels

15-25%

Efficiency drop
in peak summer

When demand is highest

45 °C

Peak ambient
temperature (KGP)

May-June every year

Rs. Crores

Annual campus
cooling cost

And growing every year

Why this problem matters:

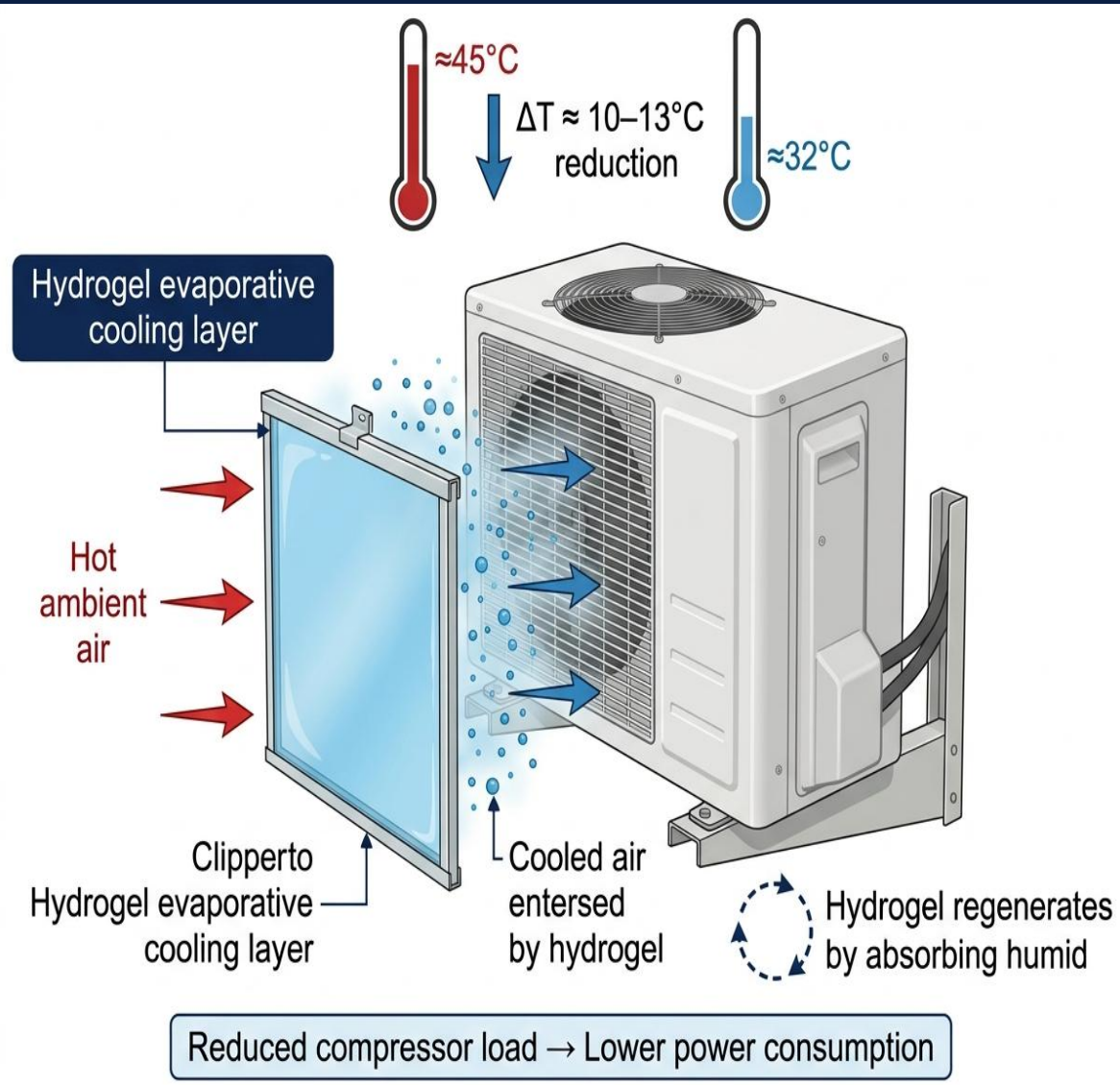
- IIT KGP's cooling is largely served by distributed split ACs — not centralised chilled water.
- Split AC efficiency degrades in hot weather: the outdoor condenser must reject heat to increasingly hot ambient air, forcing the compressor to work harder — and consume more electricity — precisely when cooling demand is highest.
- Replacing units is capital-intensive; most efficiency upgrades need extra energy, water, or maintenance.
- **Even a 5-10% improvement across 1000+ units yields substantial cumulative savings and meaningful carbon reduction.**

Why Existing Approaches Fall Short

The gap that SHEETAL fills

Replace With New Units	Continuous Water Spray	Shade Structures / Films	SHEETAL (Our Innovation)
X Rs. 30-60K per unit X Major campus disruption X Creates e-waste X Not scalable	X Needs pump + piping X Legionella / scaling risk X High water demand X Ongoing maintenance	X Marginal gain (< 3%) X Can block airflow X Structural load on wall X No active cooling	✓ Completely passive ✓ No pump / no power ✓ Retrofit in < 30 min ✓ Est. 15-20% COP gain

SHEETAL: The Proposed Innovation



What is SHEETAL?

A PAM-LiCl hydrogel panel retrofitted onto the air intake of any split AC outdoor condenser unit.

Core mechanism:

The hygroscopic hydrogel absorbs atmospheric moisture at night and releases it via evaporation during hot daytime periods, cooling the condenser inlet air by $4\text{--}10^{\circ}\text{C}$.

What makes it novel:

Applies atmospheric water harvesting + desiccant cooling principles as a zero-energy AC retrofit. No prior commercial product combines these into a passive condenser pre-cooler.

Key advantage:

Completely passive. No pump, no water supply, no external power, no structural change. Clips onto any condenser grille in under 30 minutes.

Working Mechanism: Day & Night Cycle

DAYTIME — Evaporative Cooling Cycle

1. Hot ambient air (40-45 °C, RH 25-40%) sweeps the condenser intake.
2. Air passes through the PAM-LiCl hydrogel panel.
3. LiCl lowers the vapour pressure at the gel surface ($a_u \approx 0.80$).
4. Moisture evaporates from gel into the airstream → latent heat absorbed.
5. Condenser inlet air temperature drops 4-10 °C.
6. Cooler air → better heat rejection → lower compressor head pressure.
7. Compressor power reduces → lower electricity draw → higher COP.

NIGHT — Autonomous Regeneration Cycle

1. Temperature drops (25-32 °C), relative humidity rises (60-80%).
2. Gel equilibrium RH $\approx$ 80% (for 30 wt% LiCl) → below ambient RH.
3. Atmospheric moisture flows into the gel (spontaneous adsorption).
4. PAM matrix swells, storing water for the next day's cycle.
5. No pump, no valve, no active control — fully autonomous.
6. Hydrogel is fully recharged by morning.
7. Cycle repeats: like a desert beetle, every single day.

PAM-LiCl Hydrogel: Material Selection Rationale

Water Activity (a_w)

0.75-0.85 at 30 wt% LiCl
(Conde 2004)
Equilibrium RH = 75-85%
Enables evaporation when
ambient RH < 80%

Water Uptake

> 5 g water / g polymer
(PAM hydrogel)
Thin 3 mm panel holds
enough water for 15-50 h
of continuous cooling

Evaporation Rate

$j \approx 50 \mu\text{g}/(\text{m}^2 \cdot \text{s})$
(Analytical, 45°C, RH 30%)
Surface cooling: 4-10 K
COP improvement: ~15-20%

Self-Regeneration

Absorbs moisture from
humid night air passively
Cycle life: > 500 cycles
(Li et al. 2018 ref.)
No cartridge replacement

Synthesis & Cost

Free-radical polymerisation
Low-cost: AAm + MBA + LiCl
LiCl impregnation post-gelation
Scalable to sheet format
Est. material: Rs. 200-400/unit

Analytical Model (NTU-Effectiveness + Chilton-Colburn)

Key Model Predictions

Condenser Inlet
Temp. Drop

4.4 K

Bulk air: 45 °C → 40.6 °C
(40 cm panel, u=0.35 m/s)

COP Improvement
(Estimated)

~20%

Carnot-based, T_{cond}
drop of ~5 K

Nu / Sh
(Graetz Corr.)

6.2 / 6.0

Entry-length correction
Re = 413, Pr = 0.69

Panel Endurance

15-50 h

3-10 mm PAM-LiCl
before night recharge

Governing equations (base case: L=40 cm, H=20 mm, u=0.35 m/s, T_{in}=45°C, RH=30%, PAM-LiCl 30 wt%)

$$h_{conv} = Nu * k_{air} / H = 8.42 \text{ W}/(\text{m}^2\text{K}) \quad NTU_T = h_{conv} * L / (G * c_p) = 0.42 \quad T_{out} = T_{gel} + (T_{in} - T_{gel}) * \exp(-NTU_T)$$

$$h_{mass} = Sh * D_{vap} / H = 9.05 \text{ mm/s} \quad NTU_m = h_{mass} * L / (u * H) = 0.52 \quad \omega_{out} = \omega_{gel} + (\omega_{in} - \omega_{gel}) * \exp(-NTU_m)$$

$$\omega_{gel} = a_w(w_{LiCl}) * \omega_{sat}(T_{gel}) \quad j_{evap} = \rho * h_{mass} * (\omega_{gel} - \omega_{air}) \quad Endurance = m_{water} / j_{evap}$$

Prototype Design & Experimental Validation Plan

Technology Readiness Level	
DONE	Basic principles observed
DONE	Technology concept formulated
TRL 3	Experimental PoC (next 2 weeks)
TRL 4	Lab-scale validation
TRL 5	Relevant environment test
TRL 6	Prototype demo in situ
TRL 7	System prototype operational
TRL 8	System complete & qualified
TRL 9	Deployment proven

Prototype & Validation Plan	
Phase 1 — Hydrogel Synthesis	Synthesise PAM-LiCl sheets (30 wt% LiCl). Measure water uptake, a_w, evaporation rate at 35-45 °C in lab.
Phase 2 — Bench Channel Test	Lab-scale air channel (40x20 mm). Heater + hygrometer. Measure DeltaT and j_evap. Validate vs. analytical model.
Phase 3 — Condenser Mockup	Aluminium frame + gel panel for standard 1.5T split AC (800x550 mm face). Test fit and pressure drop.
Phase 4 — Field Trial (SESE Building)	Install on 2-3 AC units. Log compressor current, power, indoor T over 2-4 weeks. Compare with controls.
Key Metrics	DeltaT_condenser, DeltaW_compressor, COP_before vs COP_after, gel recharge time, daily water balance.

How Big Can This Get?

IIT KGP Campus

~1000 split AC units

- 20% COP improvement
- 15-25% electricity saving on AC
- Meaningful carbon reduction
- Visible net-zero contribution

Indian Campuses

>1000 universities / IITs / NITs

- Millions of split AC units
- Procurement via sustainability MoU
- Replicable IIT Kharagpur model
- Central govt. green campus push

Indian Residential

300 million+ AC units in India

- World's fastest-growing AC market
- No infrastructure change needed
- Works on any split AC brand
- Affordable retrofit for all

Global South

Hot & humid tropical climates

- Maximum benefit in hot countries
- No grid power required
- Climate adaptation technology
- Aligns with SDG 7 & SDG 13

Path to Deployment

Unit Economics (Estimated)	
Material cost (per unit):	~Rs. 200-400
Frame + fitting hardware:	~Rs. 300-500
Total manufacturing cost:	~Rs. 500-900
Retail price (2x markup):	~Rs. 1,000-1,800
Annual electricity saving:	~Rs. 1,500-4,000
Simple payback period:	< 6 months
Panel lifespan:	> 3 years (est.)
ROI over 3 years:	4-10x investment

Go-to-Market Strategy
Phase 1 (0-6 months) — Campus Pilot Install on 50 AC units at IIT KGP. Publish performance data. Build credibility with real numbers.
Phase 2 (6-18 months) — Institute Network License to 10 IITs / NITs via campus sustainability procurement MoU.
Phase 3 (18-36 months) — B2B Channel Partner with AC service companies (Daikin, Voltas, Blue Star dealers) as an add-on retrofit product.
Phase 4 (3+ years) — Consumer Retail D2C via Amazon / Flipkart. Self-install kit format. Refill cartridge subscription (~Rs. 200/year).
Revenue model Product sale + annual refill subscription. Institutional bulk pricing for colleges and housing societies.

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SHEETAL is not just an idea — it is a thermodynamically grounded, low-cost, and passively scalable solution ready to begin deployment within IIT Kharagpur this academic year.